

Scribe 2 (Part 2): Questions 3–6

Question 3: Probabilistic Modeling with Deterministic Decision Rule

While the routing decision in the system is deterministic, the environment in which the route operates is not. The travel time over the road segments varies because of the changes in the amount of traffic. To account for the uncertainty in the environment, the travel time over the road segments is modeled as a random variable.

Let T_i be the travel time over the i^{th} road segment. We model the travel time over the road segments to be lognormally distributed:

$$T_i \sim \text{Lognormal}(\mu_i, \sigma_i^2)$$

This is a reasonable assumption because the travel time over the segments is always positive, and the lognormal distribution is right skewed.

While the environment in which the route operates is uncertain, the routing decision is deterministic. The routing decision is made using the A* shortest path algorithm.

The expected travel time over a route consisting of n segments is given by:

$$T_{\text{route}} = \sum_{i=1}^n T_i$$

A deadline D is set for the route. The probability of missing the deadline is estimated using the simulation:

$$P(T_{\text{route}} \geq D)$$

Additionally, we compute a theoretical upper bound using Markov's inequality:

$$P(T_{\text{route}} \geq D) \leq \frac{\mathbb{E}[T_{\text{route}}]}{D}$$

Thus, uncertainty affects travel time outcomes, but the routing strategy itself remains deterministic.

Question 4: Simulation Pipeline

The simulation pipeline models the travel process under uncertainty while maintaining a deterministic routing decision.

1. Input Generation

Random travel times for each road segment are generated using a lognormal distribution. These samples represent variability in traffic conditions.

2. Deterministic Route Selection

The system computes a fixed route using the A* algorithm. The algorithm minimizes the expected total travel time using deterministic edge weights.

3. Route Time Simulation

For each simulation run, travel times are sampled for each segment along the selected route. The total route time is computed as:

$$T_{\text{route}}^{(k)} = \sum_{i=1}^n T_i^{(k)}$$

where k denotes the simulation iteration.

4. Deadline Comparison

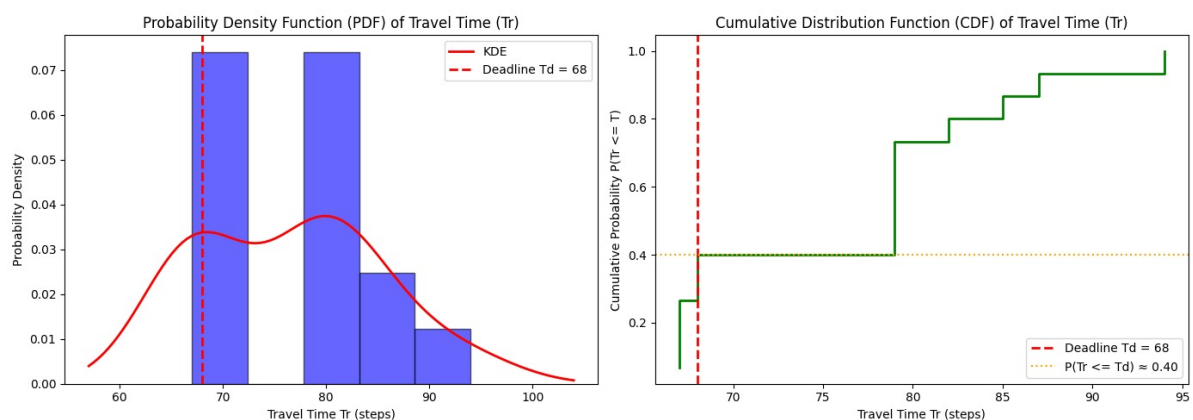
Each simulated route time is compared with the deadline D to determine whether the route arrives early or late.

5. Output Metrics

Simulation outputs are stored for statistical analysis and visualization.

Question 5: Simulation Outputs and Interpretation

Several statistical outputs are generated to analyze the behavior of the deterministic routing strategy.



Probability Density Functions (PDF)

PDF plots are used to visualize the distribution of:

- Segment travel times

- Total route travel time
- Arrival time relative to the deadline

The segment travel time distribution typically shows right-skewness, reflecting occasional traffic delays.

Cumulative Distribution Functions (CDF)

CDF plots illustrate the probability that travel time is below a certain threshold:

$$F(t) = P(T \leq t)$$

From the CDF of the total route time, we can estimate the probability of arriving before the deadline.

Arrival Relative to Deadline

We define the arrival deviation as:

$$A = T_{\text{route}} - D$$

- $A < 0$ indicates early arrival
- $A > 0$ indicates late arrival

The corresponding PDF and CDF provide insight into the reliability of the route.

Question 6: Technical Gaps and Preparation for Next Stage

The deterministic baseline is a good reference but has some drawbacks.

Simplified Traffic Modeling

The current model assumes that the segment travel time is independent. However, traffic data can have some correlation.

Fixed Routing Strategy

The A* algorithm calculates the routes based on the expected time. However, the routing does not change based on the traffic variability.

Limited Decision Flexibility

As the routing strategy is deterministic, there is no consideration of the uncertainty during the routing decisions.

Future Improvements

In Milestone 4, we plan to extend the current deterministic routing framework by incorporating randomized decision-making strategies that explicitly account for travel time uncertainty. Instead of always selecting a single fixed path based only on expected travel times, the system may probabilistically choose among multiple candidate routes depending on their reliability and risk levels.

Possible improvements include:

- **Risk-aware routing strategies:** selecting routes based not only on expected travel time but also on the probability of meeting the deadline.
- **Probabilistic path planning:** assigning probabilities to multiple feasible routes and selecting among them using randomized policies.
- **Adaptive routing under uncertainty:** dynamically updating route choices as new traffic information becomes available.

These randomized strategies can improve route reliability and reduce the probability of missing deadlines in highly uncertain traffic environments.

End of Submission